

Computer Science Series • Grade 6

Chapter 1: Computer Virus

A complete structural guide to understanding malicious software, infection types, prevention strategies, and antivirus protection tools.

 Standard Curriculum |  Cybersecurity Foundations

TEACHER TIP

Welcome the students to Grade 6 Computer Science. Ask them if their personal or family computer has ever run slowly or stopped working, and bridge that into the topic of "Biological Viruses vs Computer Viruses". Refer directly to Varsity Education's Chapter 1.

Learning Objectives

CHAPTER GOALS

By the end of this session, students will be fully equipped to master and apply these standard syllabus goals:

- i Define Computer Viruses:** Grasp what a virus is and understand its full acronym.
- i Identify Transmission Paths:** Explore exactly how viruses travel between computers.
- i Recognize Key Symptoms:** Notice immediately when a computer is showing signs of infection.
- i Explore Malware Categories:** Distinguish between regular Program Viruses, Boot Sector, FAT Viruses, Worms, Trojans, and Logic Bombs.
- i Implement Defense Tactics:** Protect computer systems using active defense strategies.
- i Leverage Antivirus Tools:** Name and run popular professional security suites.

 Cybersecurity shield protection glow concept indicating defense system

PEDAGOGY

Introduce the six clear goals of WhatsApp Image 2026-06-22 at 9.34.08 AM.jpeg. Keep the focus interactive by assuring students that they'll be cybersecurity detectives by the end of this lesson!

What is a Computer Virus?

CORE DEFINITIONS

Core Definition

A **Computer Virus** is a malicious software program designed by programmers to replicate, spread itself, and purposely damage data or software stored on a computer.

Standard Acronym (VIRUS)

V.I.R.U.S.

Vital Information Resources Under Siege

-  **Executable Nature:** It is basically an executable program written to cause disruption.
-  **File Operations:** It can actively create, move, erase, or modify private system files.
-  **Resource Eater:** It consumes massive amounts of RAM and disk storage space.
-  **Network Spreader:** Can copy itself to multiple connected devices across internet links.

TEACHER NOTE

Emphasize the standard spelling of the acronym found in Page 7 (VIRUS: Vital Information Resources Under Siege). Highlight that a virus is NOT hardware, but an execution program code written intentionally by bad actors.

Transmission & Spread

INFECTION PATHS



Email Attachments

Opening suspicious emails or media attachments (Most common method).



Removable Drives

Plugging in infected USB pen drives or floppy/flash cards from untrusted places.



Pirated Downloads

Using cracked programs, game mods, or files shared on public sites.



Insecure Surfing

Browsing compromised pages, letting systems run old software updates.

⚠ The Gold Standard of Cybersecurity: Never transfer file directories from external systems without doing a complete Antivirus Sweep first! Unchecked downloads trigger instant automatic infections.

ENGAGEMENT

Discuss Page 7 and 8 details on propagation. Inform students that simply having an infected file on a computer doesn't cause damage until it is **executed** (opened/run). This concept is vital for exams.

⚠ Warning Signs & Symptoms

INFECTION DIAGNOSTICS

- ❗ **Hanging & Freezing:** System stops responding completely or frequently hangs.
- ❗ **Unusually Slow Speed:** Simple apps take long times to launch or operate slowly.
- ❗ **Unexpected Pop-ups:** Ads or weird alerts appear on screen when offline.
- ❗ **Inaccessible Drives:** Hardware paths, hard disk blocks, or USB folders refuse to load.
- ❗ **Mysterious Data Losses:** Essential documents or program files disappear or end up hidden.
- ❗ **App Malfunctions:** Installed programs crash randomly without showing a clear error.



? Fact vs Fiction

COMMON MYTHS CLEARED

Write-Protected Disks

Viruses **CANNOT** write onto protected structures.

Read-only physical storage like **CD-ROMs** or **DVD-ROMs** protect content. Because a computer drive cannot overwrite them, a virus cannot attach itself to them.

Physical Hardware Protection

Viruses **CANNOT** damage hardware components.

A computer virus is made up of computer programs (software code). It can corrupt files and software, but it **cannot physically damage** hardware pieces like chips, screens, or keyboards.

Exam Point: "Can a virus burn your CPU physically?" **No!** Software code damages software systems, not physical boards.

Types of Malware: System & Files

CATEGORIZING ENEMIES

Program Viruses

Target: Application Extensions

These target program execution directories. They attach themselves to files with extension types like **.com**, **.exe**, **.sys**, **.dll**. Once run, the payload starts executing.

Boot Sector Virus

Target: MBR (Master Boot Record)

These infect the drive's boot block sector. It loads directly into computer memory during system startup, before any software protections can even load up.

FAT Virus

Target: File Allocation Table

The FAT stores the directory paths of all saved data. By attacking this index table, these viruses break the system's ability to find where files are stored, causing massive file errors.

Worms, Trojans & Logic Bombs

SOPHISTICATED THREATS

Computer Worms

Independent Self-Replicators

Unlike standard viruses, worms **do not need** to attach to an existing file. They are separate files that replicate on their own over networks, slowing systems down dramatically.

Trojan Horse

Decoy Programs

Disguised as useful programs (like games or helper tools), Trojans hide their malicious intent inside. They do not self-replicate, but they open secret access backdoors for hackers.

Logic Bombs

Time/Trigger Attacks

These secret sections of code lie quiet inside a system until a specific condition is met, like a calendar date (e.g., Friday the 13th) or a user logging in. Once triggered, they explode into action.

Spyware & General Malware

SPY BOTS & TRACKING SOFTWARE

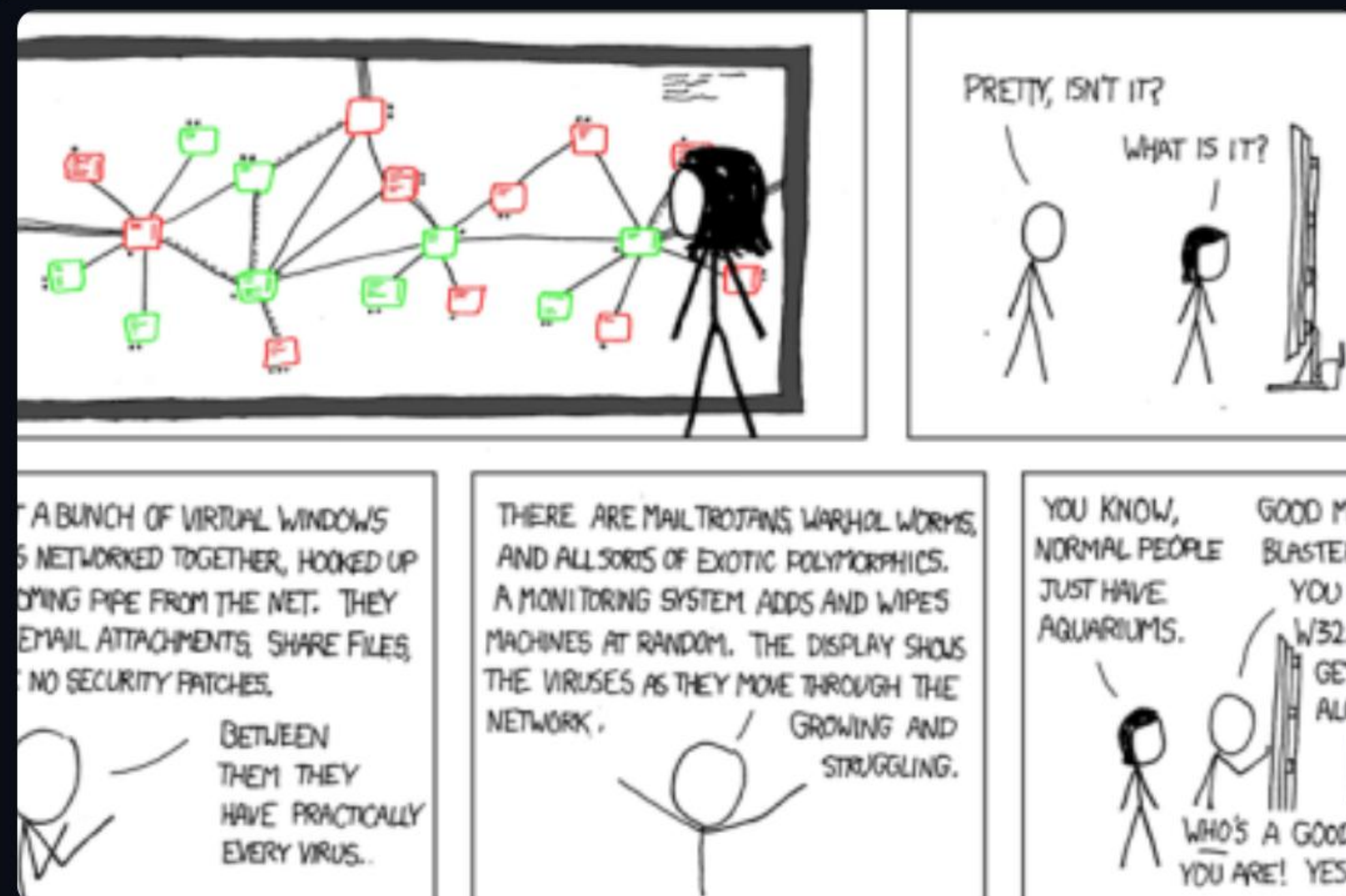
What is Spyware?

Also known as **Spybots** or **Tracking Software**. It runs silently in the background, spying on your computer habits, browsing history, and passwords, and sends this data back to hackers.

Examples: **Spy anytime**, **n-case**, **tracking cookies**, **adware**.

Malware (Malicious Software)

An umbrella term that stands for "**Malicious Software**". This term covers all hostile programs, including standard viruses, spyware, tracking worms, and hidden trojan horses.



Active Defense Strategies

SYSTEM PROTECTION

Keep these safety practices active to secure your system folders from infection:

- i Turn On Firewall:** Keep your network firewall turned on to filter incoming internet connections.
- i Block Suspicious Links:** Avoid sites that promise illegal downloads, free hacks, or mods.
- i Set Automatic Updates:** Keep your OS up to date with the latest security fixes.
- i Verify Attachments:** Do not open email attachments from unknown senders.
- i Use Pop-up Blockers:** Turn on browser blockers to stop dangerous surprise alerts.
- i Clear Cache:** Regularly clear browser caches and local tracking histories.

 cyber security secure dashboard lock system

✓ What is it?

An **Antivirus** is a security program used to scan, identify, block, and clean malicious programs from your system files.

Famous Professional Tools

✓ Kaspersky Anti-Virus

✓ McAfee Security

✓ AVG Internet Security

✓ Norton Security

Essential Protection Checklist:

-  Perform routine storage drive sweeps.
-  Update virus definition files daily.
-  Isolate infected systems instantly.
-  Verify removable USB keys before running files.

Interactive Activity

SPOT THE ANTIVIRUS LOGOS

Look at the brands below. Which of these are real Antivirus programs? Point out the correct options!



Kaspersky Security



Windows Clouds



AVG Security



Action Game App



ESET NOD32



Logic Spyware



Avast Antivirus



Zip Unpacker



McAfee Personal

File Editor Tool

✔ Answer Check: Kaspersky, AVG, ESET NOD32, Avast, and McAfee are correct!

ACTIVITY

Based on the Activity on Page 11 of the textbook. Encourage your students to raise their hands and spot the right security programs on screen.

Revision Notes

IMPORTANT HIGHLIGHTS

Must-Know Facts

-  **VIRUS:** Vital Information Resources Under Siege.
-  **Malicious Code:** Written intentionally by human coders.
-  **Target Files:** Program execution paths like **.exe, .com**.
-  **Worms:** Replicate on their own without needing host files.

Defending Systems

-  Do not execute suspicious system files.
-  Keep firewalls and security settings turned on.
-  Use updated Antivirus tools (McAfee, Avast, AVG).
-  Avoid untrusted file transfers or unverified downloads.

TEACHER TIP

These revision notes are great for review sessions right before the exam. Keep definitions clear and simple for Class/Grade 6.

Practice Questions

ASSESS YOUR UNDERSTANDING

Q1: What does the 'V' stand for in VIRUS?

A) Virtual

B) Vital (Correct)

C) Verify

Q2: Which software replicates without needing a host program file?

A) Worm (Correct)

B) Logic Bomb

C) Trojan Horse

QUIZ TIME

Run these questions with the class. Encourage students to explain **why** option B is correct for Q1, and option A is correct for Q2 based on the chapter's definitions.

Session Summary

WRAP UP

Here is a quick summary of what we learned today:

We defined computer viruses, learned the VIRUS acronym, and looked at different malware categories like Program, Boot Sector, and FAT Viruses, as well as Worms, Trojans, and Logic Bombs.

We studied system defense tips, including pop-up blockers, firewall sweeps, OS updates, and using trustworthy Antivirus software to clean infected drives.



Mastery Achieved!

You are now ready to keep your computers safe from viruses and help protect your family's devices too!

WRAP UP

Congratulate the students on completing Chapter 1. Introduce the take-home tasks and question banks shown on the next page.

Image Sources



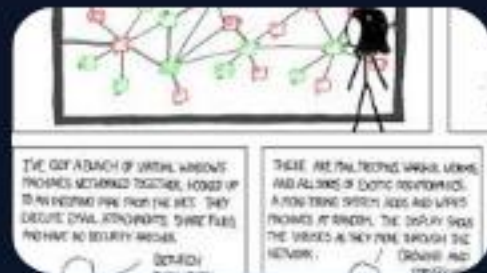
https://static.vecteezy.com/system/resources/previews/028/627/269/non_2x/shield-polygonal-symbol-data-protection-concept-cyber-security-concept-with-glowing-low-poly-shield-on-geometric-background-antivirus-concept-cyber-shield-internet-safety-design-illustration-vector.jpg

Source: www.vecteezy.com



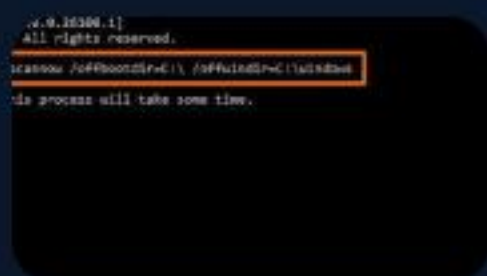
Thumbnail for https://mycomputerworks.com/wp-content/uploads/2017/08/shutterstock_127762841-min.jpg

Source: mycomputerworks.com



<https://jon.oberheide.org/images/2008/03/network.png>

Source: jon.oberheide.org



https://www.avast.com/hs-fs/hubfs/New_Avast_Academy/how_to_fix_corrupted_files_on_windows_using_sfc_and_dism_academy/Avast-repair-corrupted-files-with-system-file-checker-screenshot-04.jpg?width=1000&height=425&name=Avast-repair-corrupted-files-with-system-file-checker-screenshot-04.jpg

Source: www.avast.com